Inequalities

Q) $x \leq -4$

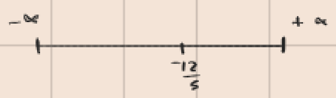
$= (-\infty, -4]$

Q) $3 + 7x \leq 2x - 9$

$7x - 2x \leq -9 - 3$

$5x \leq -12$

$x \leq \frac{-12}{5}$



$= (-\infty, -\frac{12}{5}]$

Q) $3x - 2 \leq 7$

$3x \leq 7 + 2$

$3x \leq 9$

$x \leq \cancel{\frac{9}{3}}\,3$

$x \leq 3$

$= (-\infty, 3]$

Q) $7 - 2x \leq 5 < 12 - 3x$

$7 - 2x \leq 5$

$-2x \leq 5 - 7$

$-\cancel{2}x \leq -\cancel{2}$

$x \geq 1$

$5 < 12 - 3x$

$-3x > 5 - 12$

$-3x > -7$

$x < \frac{7}{3}$

$1 \leq x < \frac{7}{3}$

$= [1, \frac{7}{3})$

Q) $7 \leq 2-5x < 9$

$7 \leq 2-5x$

$7-2 \leq -5x$

$5 \leq -5x$

$\frac{5}{-5} \geq x$

$x \leq -1$

$2-5x < 9$

$-5x < 9-2$

$-5x < 7$

$x > -\frac{7}{5}$

$-\frac{7}{5} < x \leq -1$

$= \left(-\frac{7}{5}, -1\right]$

Q) $3 \leq 2x+1 < 9$

$3 \leq 2x+1$

$3-1 \leq 2x$

$2 \leq 2x$

$\frac{2}{2} \leq x$

$x \geq 1$

$2x+1 < 9$

$2x < 9-1$

$2x < 8$

$x < \frac{\cancel{8}\,4}{\cancel{2}\,1}$

$x < 4$

$1 \leq x < 4$

$[1, 4)$

Q) $x^2 - 5x + 6 > 0$

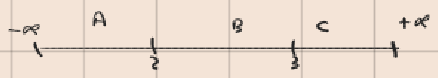
$x^2 - (3+2)x + 6 > 0$

$x^2 - 3x - 2x + 6 > 0$

$x(x-3) - 2(x-3) > 0$

$(x-3)(x-2)$

$x = 2$ $\quad$ $x = 3$



$x < 2$	1	$(1)^2 - 5(1) + 6 = 1 - 5 + 6 = -4 + 6 = 2$ $\quad +$
$2 < x < 3$	1.5	$(\frac{3}{2})^2 - 5(\frac{3}{2}) + 6 = \frac{9}{4} - \frac{15}{2} + 6 = -\frac{1}{4}$ $\quad -$
$x > 3$	4	$= 2$ $\quad +$

$x \in (-\infty, 2) \cup (3, \infty)$

Q) $3x^2 - 7x - 6 < 0$

$3x^2 - (9-2)x - 6 < 0$

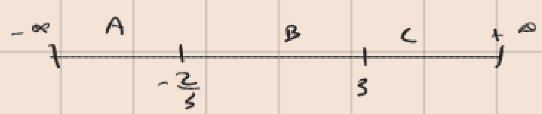
$3x^2 - 9x + 2x - 6 < 0$

$3x(x-3) + 2(x-3) < 0$

$(3x+2)(x-3) < 0$

$3x \; 2 = 0$

$x = -\frac{2}{3}$ $\quad$ $x = 3$



$x < -\frac{2}{3}$	-1	$-$
$-\frac{2}{3} < x < 3$	0	$+$
$x > 3$	4	$-$

$x \in (-\frac{2}{3}, +3$

Q) $-2x^2 + 5x + 3 \leq 0$

$-(2x^2 - 5x - 3) \leq 0$

$2x^2 - (6-1)x - 3 \geq 0$

$2x^2 - 6x + x - 3 \geq 0$

$2x(x-3) + 1(x-3) \geq 0$

$(2x+1) \quad (x-3)$

$x = \frac{-1}{2}$ $\qquad$ $x = 3$

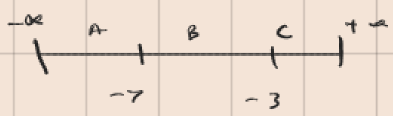
$x < \frac{-1}{2}$	-1	$4 \geq 0$	$+$
$\frac{-1}{2} < x < 3$	0	$-3 \geq 0$	$-$
$x > 3$	4	$9 \geq 0$	$+$

$x \in (-\infty, \frac{-1}{2}] \cup [3, \infty]$

Q) $\dfrac{x+7}{x+3} > 0$

$x + 7 = 0$ $\qquad$ $x + 3 = 0$

$x = -7$ $\qquad$ $x = -3$



$x < -7$	-8	$\frac{1}{5} > 0$	$+$
$-7 < x < -3$	-5	$-1 > 0$	$-$
$x > -3$	-1	$3 > 0$	$+$

$x \in (-\infty, -7) \cup (-3, \infty)$

Q) $\dfrac{x+1}{x-2} \le 3$ ✗

$x+1 = 3$ $\quad$ $x-2 = 3$

$x = 3-1$ $\quad$ $x = 3+2$

$x = 2$ $\quad$ $x = 5$

$-\infty$ A B C $+\infty$; 2, 5

$x < 2$	1	$-2 \le 3$	+
$2 < x < 5$	3	$4 \le 3$	−
$x > 5$	6	$\frac{7}{4} \le 3$	+

$x \in (-\infty, 2) \cup [5, +\infty)$

Q) $\dfrac{x+1}{x-2} \le 3$

$\dfrac{x+1}{x-2} - 3 \le 0$

$\dfrac{x+1 - 3(x-2)}{x-2} \le 0$

$\dfrac{x+1 -3x +6}{x-2} \le 0$

$\dfrac{-2x + 7}{x-2} \le 0$

$-2x + 7 \le 0$ $\quad$ $x-2 \le 0$

$-2x \le -7$ $\quad$ $x \le +2$

$x \geq \frac{+7}{+2}$

$x \geq \frac{7}{2}$ $\quad$ $x \le +2$

$-\infty$ A B C $+\infty$; 2, $\frac{7}{2}$

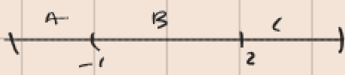
$x < 2$	1	$-5 \le 0$	+
$2 < x < \frac{7}{2}$	3	$1 \le 0$	−
$x > \frac{7}{2}$	4	$-\frac{1}{2} \le 0$	+

$x \in (-\infty, 2) \cup [\frac{7}{2}, +\infty)$

Q) $\dfrac{x+1}{x-2} \geq 0$

$x+1 \geq 0$ $\quad$ $x-2 \geq 0$

$x \geq -1$ $\quad$ $x \geq +2$



$x < -1$	-2	$\frac{1}{4} \geq 0$	$+$
$-1 < x < 2$	0	$-\frac{1}{2} \geq 0$	$-$
$x > 2$	3	$4 \geq 0$	$+$

$x \in (-\infty, -1] \cup (2, +\infty)$

Q) $\dfrac{x^2-4}{x+1} > 0$

$\dfrac{x^2-2^2}{x+1}$

$\dfrac{(x+2)(x-2)}{x+1} > 0$

$x+2 > 0$ $\quad$ $x-2 > 0$ $\quad$ $x+1 > 0$

$x > -2$ $\quad$ $x > 2$ $\quad$ $x > -1$

$x < -2$	-3	$-\frac{5}{2} > 0$	$-$
$-2 < x < -1$	-1.5	$\frac{7}{2} > 0$	$+$
$-1 < x < 2$	0	$-4 > 0$	$-$
$x > 2$	3	$\frac{5}{4} > 0$	$+$

$x \in (-2, -1) \cup (2, \infty)$

1) $|x| < a$

$-a < x < a$

2) $|x| \leq a$

$-a \leq x \leq a$

3) $|x| > a$

$x > a$ or $x < -a$

4) $|x| \geq a$

$x \geq a$ or $x \leq -a$

Q) $\left|\dfrac{2x+2}{4}\right| + 4 \leq 5$

$\left|\dfrac{2x+2}{4}\right| \leq 5-1$

$\dfrac{2x+2}{4} \leq \pm 1$

$\dfrac{2x+2}{4} \leq +1$

$2x+2 \leq 4$

$2x \leq 4-2$

$2x \leq 2$

$x \leq 1$

$\dfrac{2x+2}{4} \leq -1$

$2x+2 \leq -4$

$2x \leq -4-2$

$2x \leq -6$

$x \leq -3$

$-1 \leq \dfrac{2x+2}{4} \leq 1$

$\dfrac{2x+2}{4} \geq -1$

$2x+2 \geq -4$

$2x \geq -4-2$

$x \geq \dfrac{-6}{2}$

$x \geq -3$

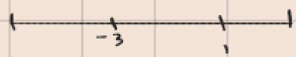
$\dfrac{2x+2}{4} \leq 1$

$2x+2 \leq 4$

$2x \leq 4-2$

$2x \leq 2$

$x \leq 1$



$[-3, 1]$

$x < -3$	-4	$\frac{3}{2} + 4 \leq 5$	$-$
$-3 < x < 1$	0	$\frac{1}{2} + 4 \leq 5$	$+$
$x > 1$	2	$\frac{3}{2} + 4 \leq 5$	$-$

$x \in [-3, 1]$

Functions

+ve meas upward faid

Q) $\therefore 2|x+2|+1$ Find the vertex

$x+2=0$

$x=-2$

$y = 2|-2+2|+1$

$y = 1$

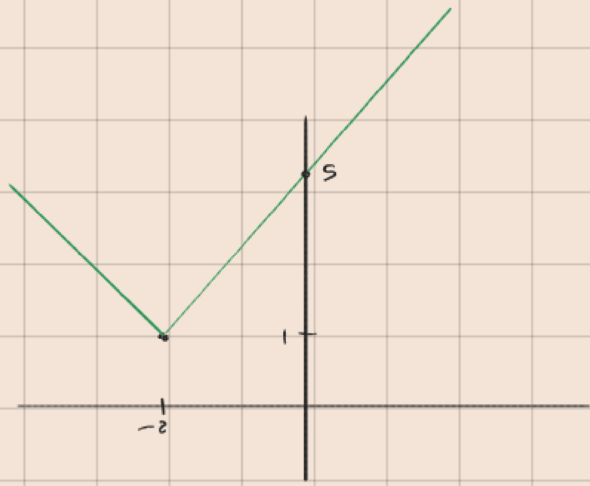
$x=0$

$y = 2(+2)+1$

$y = 4+1$

$y = 5$ y inte = $(0,5)$

vertex $(-2,1)$

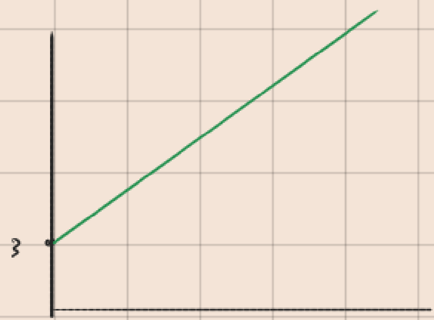


Domain $(-\infty, \infty)$

Range $[1, \infty)$

Q) $f(x) = 2x+3$

$y = mx + c$



For Quadratic

ax^2+bx+c

(smily face)

Domain $(-\infty, \infty)$

$a>0$ - $[k, \infty)$

$-ax^2+bx+c$

(frown)

Domain: $(-\infty, \infty)$

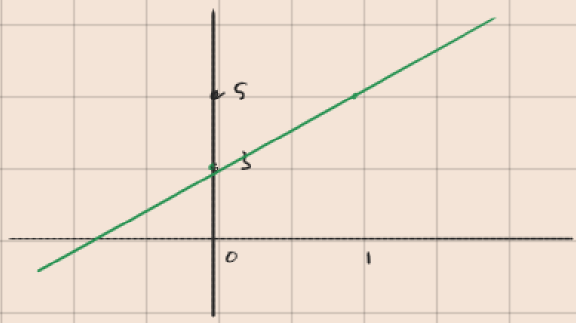
$a<0$: $(-\infty, k] \cup (-\infty, k]$

vertex: (x,y)

$x = \frac{-b}{2a}$

Q) $f(x) = 2x + 3$

x	0	1
y	3	5



Domain = $(-\infty, \infty)$

range = $(-\infty, \infty)$

Q) $f(x) = 5$

Domain $(-\infty, \infty)$

range = $\{5\}$

Q) $y = x^2 - 6x + 5$

vertex $(3, -4)$

$x = \frac{-(-6)}{2(1)}$

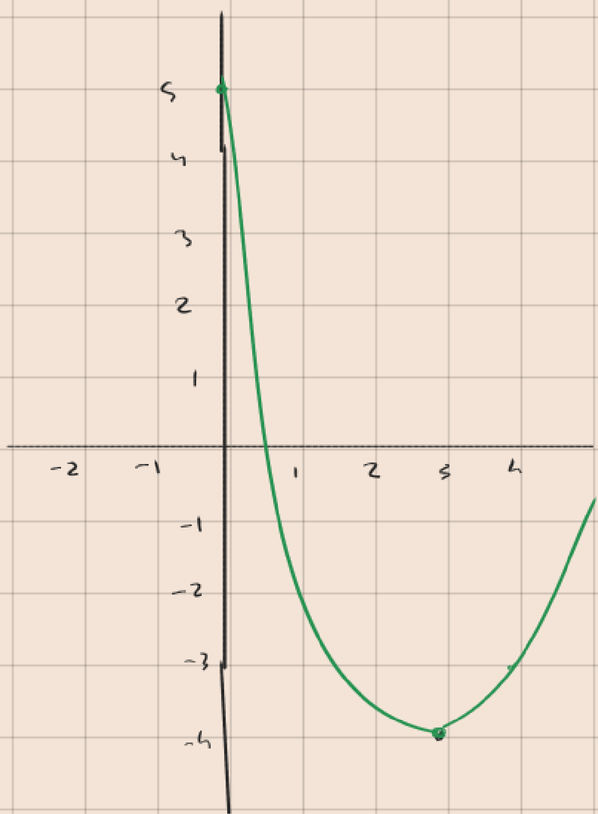
$y = 3^2 - 6(3) + 5$

$y = -4$

$x = 3$

Domain $(-\infty, \infty)$

Range $[-4, +\infty)$

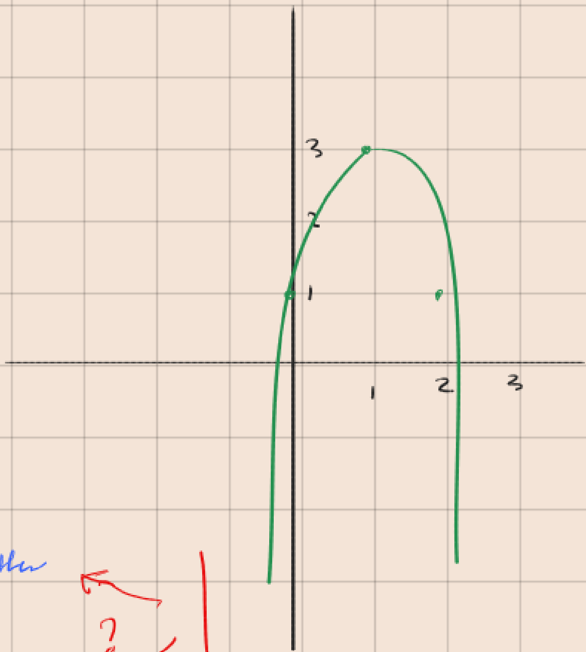


e) $-2x^2 + 4x + 1$

vertex $(1, 3)$

$x = \frac{-(4)}{2(-2)}$

$x = 1$ $\quad y = 3$



y int $= 1$ Domain $(-\infty, +\infty)$

Range $[3, -\infty)$ better

?

Range $= (-\infty, 3] \cup [3, -\infty)$

$x + h$

$9x + 6$

$\left(-\frac{2}{3}, 0\right)$

0

Q) $\lim_{x \to 2} \frac{x-2}{\sqrt{x+2}}$

$= \frac{2-2}{\sqrt{2+2}}$

$= \frac{0}{4} = 0$

Q) $\lim_{x \to 1} \frac{x^3-1}{x-1}$

$= \frac{x^3-1^3}{x-1}$

$= \frac{\cancel{(x-1)}(x^2+x+1)}{\cancel{(x-1)}}$

$= 1+1+1$

E

$= 3$

Q) $\lim_{x \to 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right)$

$= \left(\frac{1}{1-1} - \frac{3}{1-1} \right)$

$= \infty - \infty$ ∴ undeterminate form

$= \frac{1}{(1-x)} - \frac{3}{(1-x)(1+x+x^2)}$

$$= \frac{(1+x+x^2) - 3}{(1-x)(1+x+x^2)}$$

$$= \frac{x^2+x-2}{(1-x)(1+x+x^2)}$$

$$= \frac{x^2+(2-1)x-2}{(1-x)(1+x+x^2)}$$

$$= \frac{x^2+2x-x-2}{x(x+2)-1(x+2)}$$

$$= \frac{(x-1)(x+2)}{(1-x)(1+x+x^2)}$$

$$= \frac{-\cancel{(1-x)}(x+2)}{\cancel{(1-x)}(1+x+x^2)}$$

$$= \frac{-(x+2)}{(1+x+x^2)}$$

$$= \frac{-(1+2)}{3}$$

$$= -\frac{3}{3}$$

$$= -1$$

Q) $\lim_{x \to 0} \frac{\operatorname{cosec} x - \cot x}{x}$

$= \frac{1}{\sin x} - \frac{\cos x}{\sin x} \div x$

$= \frac{1 - \cos x}{x \cdot \sin x} \times \frac{1 + \cos x}{1 + \cos x}$

$= \frac{1^2 - \cos^2 x}{(x \sin x)(1 + \cos x)}$

$= \frac{\sin^2 x}{x \cdot \cancel{\sin x}\,(1 + \cos x}$

$= \frac{\sin x}{x} \cdot \frac{1}{1 + \cos x}$ $\therefore \frac{\sin x}{x} = 1$

$= 1 \times \frac{1}{1 + \cos x}$

$= \frac{1}{1 + \cos(0)}$

$= \frac{1}{1+1}$

$= \frac{1}{2}$

Q) $\lim_{x \to 0} \dfrac{1-\cos x}{x^2}$

$$\frac{1-\cos x}{x^2} \times \frac{1+\cos x}{1+\cos x}$$

$$= \frac{1^2 - \cos^2 x}{x^2 \cdot (1+\cos x)}$$

$$= \frac{\sin^2 x}{x^2} \times \frac{1}{1+\cos x}$$

$$= \frac{1}{2}$$

Q) $\lim_{y \to x} \dfrac{y^{\frac{2}{3}} - x^{2/3}}{y-x}$

$$\frac{(y^{\frac{1}{3}})^2 - (x^{\frac{1}{3}})^2}{(y-x)}$$

$$= \frac{(y^{\frac{1}{3}} - x^{\frac{1}{3}})(y^{\frac{1}{3}} + y^{\frac{1}{3}})}{(y^{\frac{1}{3}})^3 - (x^{\frac{1}{3}})^3}$$

$$= \frac{\cancel{(y^{\frac{1}{3}} - x^{\frac{1}{3}})}(y^{\frac{1}{3}} + x^{\frac{1}{3}})}{\cancel{(y^{\frac{1}{3}} - x^{\frac{1}{3}})}(y^{\frac{2}{3}} + y^{\frac{1}{3}}x^{\frac{1}{3}} + x^{\frac{2}{3}})}$$

$$= \frac{y^{\frac{1}{3}} + x^{\frac{1}{3}}}{y^{\frac{2}{3}} + y^{\frac{1}{3}} \times x^{\frac{1}{3}} + x^{\frac{2}{3}}}$$

$x^2 \times x^2$

$$= \frac{2x^{\frac{1}{3}}}{2x^{\frac{2}{3}} + x^{\frac{2}{3}}}$$

$\frac{1}{3} + \frac{1}{3}$

$$= \frac{2x^{\frac{1}{3}}}{3x^{\frac{2}{3}}}$$

$\therefore \frac{1}{3} - \frac{2}{3}$

$-\frac{1}{3}$

$$= \frac{2}{3x^{\frac{1}{3}}}$$

$\theta)$ $\dfrac{\tan x\,(\sin x)}{\sin x}$

$\sin x = t$

$\dfrac{\tan(t)}{t} = 1$

$\theta)$ $\dfrac{\sqrt{x^2+1}}{x+1}$?

$\dfrac{\sqrt{x^2\left(1+\frac{1}{x^2}\right)}}{x+1}$

$\theta)$ $\lim\limits_{x\to\infty} \dfrac{4x^3-2x^2+1}{3x^3-5}$

$= \dfrac{\cancel{x^3}\left(4-\frac{2}{x}+\frac{1}{x^3}\right)}{x^3\left(3-\frac{5}{x^3}\right.}$

$= \dfrac{4}{3}$

$\theta)$ $\lim\limits_{x\to\infty} \left[\dfrac{x^2}{x+1} - \dfrac{x^2}{x+3}\right]$

$\dfrac{x^2(x+3) - x^2(x+1)}{(x+1)(x+3)}$

$\dfrac{\cancel{x^3}+3x^2 - \cancel{x^3} - x^2}{x^2+3x+x+3}$

$\dfrac{+2x^2}{x^2+4x+3}$

$\dfrac{x^2}{x^2}\;\dfrac{+2}{1+}$

$= 2$

Q) $\lim_{x \to -2^-} \dfrac{x^2 + 2x - 2}{x^2 - 4}$ $\qquad \dfrac{-}{-\;+}$

$\dfrac{x^2 + 2x - 2}{x^2 + 4}$ at $x = -2$ $= \infty$

$\dfrac{x^2 + 2x - 2}{(x-2)(x+2)} = \dfrac{-\text{value}}{(-\text{val}) \times (-\text{val})} = -\infty$

Q) $\lim_{x \to \infty} \left(1 + \frac{2}{x}\right)^x$ $\qquad 1 + \frac{1}{x}$

$\left[\left(1 + \frac{2}{x}\right)^{\frac{x}{2}}\right]^2$ $\qquad x = 2n$

$\left(1 + \frac{\cancel{2}}{\cancel{2}n}\right)^{\frac{\cancel{2}n}{\cancel{2}}}$

$\left(1 + \frac{1}{n}\right)^n$

$(e)^2$

$= e^2$

Q) $f(x) = \begin{cases} x^2 + 3 & x < 1 \\ x + 1 & x > 1 \end{cases}$

left hand limit $\lim_{x \to 1^-}$	right hand
$1^2 + 3 = 4$	$1 + 1 = 2$

Q) $\begin{cases} 3 & x \leq -2 \\ -\frac{1}{2}x^2 & -2 < x \leq 2 \\ 3 & x \geqslant 2 \end{cases}$

$x = -2$

left

$= 3$

$y = 3$

right

$-\frac{1}{2}(-2)^2$

$= -2$

$y = -2$

$x = 2$

left

$-\frac{1}{2}(2)^2$

$= -2$

righ

$= 3$

Q) $\begin{cases} x - 2 & x \leq 1 \\ ax^2 & x > -1 \end{cases}$ Find the value of a

left

$1 - 2 = -1$

right

$ax^2 = -1$

$a(-1)^2 = -1$

$a = -1$

Q) $f(x) = \begin{cases} \frac{1}{x+2} & x < -2 \\ x^2 - 5 & -2 < x < 3 \\ \sqrt{x+3} & x > 3 \end{cases}$

$x = -2$

Left

$\frac{1}{-2+2} = -\infty$

righ

$(-2)^2 - 5$

$= -1$

$x = 3$

left

$(3)^2 - 5$

$= 4$

righ

$\sqrt{6}$

Practice Shit

Limits

Q1) $\lim_{x \to 0} \dfrac{\sin ax}{\sin bx}$

$\dfrac{\sin ax}{\sin bx} \times \dfrac{ax \cdot bx}{ax \cdot bx}$

$\dfrac{\sin ax}{ax} \times \dfrac{bx}{\sin bx} \times \dfrac{ax}{bx}$

$\left(\dfrac{\sin ax}{ax} \div \dfrac{\sin bx}{bx}\right) \times \dfrac{ax}{bx}$

$(1 \div 1) \times \dfrac{ax}{bx}$

$\dfrac{a\cancel{x}}{b\cancel{x}}$

$= \dfrac{a}{b}$

Q2) $\lim_{x \to 0} x \cdot \sin\left(\frac{1}{x}\right)$

$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$

$0 \leq \left|\sin\left(\frac{1}{x}\right)\right| \leq 1$

$0 \leq |x| \cdot \left|\sin\left(\frac{1}{x}\right)\right| \leq |x|$

$0 \leq y \leq |x|$

$\lim_{x \to 0} = 0$

$\lim_{x \to 0}$ $y \leq |x|$

$y \leq 0$

$y = 0$

Ans $= 0$

Q3) $\lim_{x \to \infty} \left(1 - \frac{1}{x}\right)^x$

$\left(1 + \frac{1}{x}\right)^x = e$

$\left(1 - \frac{1}{x}\right)^x = e^{-1}$

$(1 + x$

$\left(1 - \frac{1}{x}\right)^x$

$\left(\frac{x-1}{x}\right)^x$

$\left(\frac{x}{x-1}\right)^{-x}$

$= \frac{1}{\left(\frac{x-1+1}{x-1}\right)^x}$

$x \cancel{-1+1}$
$= x$

$= \frac{1}{\left(\frac{x-1}{x-1} + \frac{1}{x-1}\right)^x}$

$= \frac{1}{\left(1 + \frac{1}{x-1}\right)^x}$

$= \frac{1}{\left(1 + \frac{1}{x-1}\right)^{x-1} \cdot \left(1 + \frac{1}{x-1}\right)^{+1}}$

$= \frac{1}{e \cdot 1}$

$= \frac{1}{e}$ or e^{-1}

$\lim_{x \to \infty} \left(1 - \frac{1}{x}\right)^x$

$\left[\left(1 - \frac{1}{x}\right)^{-x}\right]^{-1}$

$\left[\left(1 + \left(\frac{1}{-x}\right)^{-x}\right)\right]^{-1}$

$-x = n$

$\left[\left(1 + \frac{1}{n}\right)^n\right]^{-1}$

$(e)^{-1}$

Q4) $\lim_{x \to \infty} \left(\frac{x}{1+x} \right)^x$

$$= \left(\frac{x - 1 + 1}{1+x} \right)^x$$

$$\left(\frac{\cancel{x+1}}{\cancel{1+x}} - \frac{1}{1+x} \right)^x$$

$$= \left(1 - \frac{1}{1+x} \right)^x$$

$$\left(1 - \frac{1}{1+x} \right)^{x+1} \cdot \left(1 - \frac{1}{1+x} \right)^{-1}$$

$$= e \cdot (1-0)^{-1}$$

$$= e \cdot 1^{-1}$$

$$= e$$

$\lim_{x \to \infty} \left(1 + \frac{1}{x} \right)^x = e$

$\lim_{x \to \infty} \left(1 - \frac{1}{x} \right)^x = e^{-1}$

$\lim_{x \to 0} (1 + x)^{1/x} = e$

Q5) $\lim_{x \to \infty} \frac{x^4 - 2x^2 + 6}{x^2 + 7}$

$$= \frac{\cancel{x^4} \left(1 - \frac{2}{x^2} + \frac{6}{x^4} \right)}{\cancel{x^4} \left(\frac{\cancel{x^2}\, 1}{\cancel{x^4}\, x^2} + \frac{7}{x^4} \right.}$$

$$= \frac{1 \, (1 - 0 + 0)}{0}$$

$$= \infty$$

Q6)

Q6) $\lim_{x \to \infty} \left(x - \sqrt{x^2 - a^2} \right)$

$$x - \sqrt{x^2-a^2} \times \frac{x + \sqrt{x^2-a^2}}{x + \sqrt{x^2-a^2}}$$

$$\frac{(x - \sqrt{x^2-a^2})(x + \sqrt{x^2-a^2})}{x + \sqrt{x^2-a^2}}$$

$$= \frac{(x^2 - x^2 - a^2)}{x + \sqrt{x^2-a^2}}$$

$$= \frac{a^2}{\infty} = 0$$

Q7) $\lim_{x \to +2^-} \frac{\sqrt{4-x^2}}{\sqrt{6-5x+x^2}}$

$6 - 5(2) + (2)^2$ $\qquad \frac{0}{0}$ not possible

$6 - 10 + 4 = 0$

$$= \frac{4-x^2}{x^2-5x+6}$$

$$= \frac{2^2 - x^2}{x^2-(3+2)x+6}$$

$$= \frac{(2-x)(2+x)}{x^2-3x+2x+6}$$

$$= \frac{(2-x)(2+x)}{x(x-3)-2(x-3)}$$

$$= \frac{(2-x)(2+x)}{(x-2)(x-3)}$$

$$\frac{(2-x)(2+x)}{(x-2)(x-3)}$$

$$\frac{-[(\cancel{x-2})(-2-x)]}{(\cancel{x-2})(x-3)}$$

$$= -\frac{(-2-x)}{x-3}$$

$$= \frac{(-2-(2))}{2-3}$$

$$= \sqrt{\frac{-4}{-1}}$$

$= 2$

Q8) $\lim_{x \to \pm\infty} \dfrac{5x^3 + 3x^2 - 1}{-4x^4 + x}$

$$\frac{x^4\left(\frac{5x^3}{x^4} + \frac{3x^2}{x^4} - \frac{1}{x^4}\right)}{x^4\left(-4 + \frac{x}{x^4}\right)}$$

$$= \frac{x^4}{x^4} \cdot \frac{(0 + 0 - 0)}{-4 + 0}$$

$$= \frac{0}{-4} = 0$$

Q9) $\lim_{x \to \infty} \dfrac{x^3+1}{x^{\frac{3}{2}}}$

$$= \infty$$

Q10) $\lim_{x \to \infty} \dfrac{x + \sin x}{x}$

when $x \to \infty$ $\dfrac{\sin x}{x} = 0$

$x \to 0$ $\dfrac{\sin x}{x} = 1$

$$\frac{x}{x} + \frac{\sin x}{x}$$

$$= 1 + 0$$

$$= 1$$

Qno 11) $f(x) = \begin{cases} x^2 - 1 & x \leq 2 \\ \sqrt{x+7} & x > 2 \end{cases}$

find the limit $f(x)$ as $x \to$

left

$x^2 - 1$

$4 - 1 = 3$

right

$x+7$

$\sqrt{2+7} = 3$

Q12) $f(x) = \begin{cases} \cos x & x \leq 0 \\ 1 - x & x > 0 \end{cases}$

find $\lim\limits_{x \to 0} f(x)$

left

$\cos(0) = 1$

right

$1 - 0 = 1$

same

Q13) $\lim\limits_{x \to 0^-} \dfrac{x}{x - |x|}$

$$\frac{x}{x - (-(x))}$$

$$\frac{x}{x + x}$$

$$= \frac{x}{2x}$$

$$= \frac{1}{2}$$

Q) $\lim_{x \to 1} \dfrac{x^3 + x^2 - 5x + 3}{x^3 - 3x + 2}$?

Q15) $g(t) = \begin{cases} t - 2 & t < 0 \\ t^2 & 0 \leq t \leq 2 \\ 2t & t > 0 \end{cases}$

a) $\lim_{t \to 0}$ Does not exit

left	right	limit
$t - 2$	$2t = 0$	$t^2 = 0$
$0 - 2 = -2$	$2(0) = 0$	$(0)^2 = 0$

b) $\lim_{t \to 1}$ t^2

$(1)^2 = 1$

c) $\lim_{t \to 2}$ $(2)^2 = 4$

Continuity

Qno1) Discuss the continuity of $x - |x|$ at $x = 1$

?

Q2) show that the function $f : R \longrightarrow R$

defined by

$$f(x) = \begin{cases} x & x \text{ is irrational} \\ 1 - x & x \text{ is irrational} \end{cases}$$

to see where it is continous, we set the two

piece wise together

$$x = 1 - x$$

$$2x = 1$$

$$x = \frac{1}{2}$$

it is only continous at $x = \frac{1}{2}$

Q3) find c such that the function

$$f(x) = \begin{cases} \dfrac{1-\sqrt{x}}{x-1} & 0 \le x < 1 \\ c & x = 1 \end{cases}$$

$$\lim_{x \to 1}$$

left

$$\frac{1-\sqrt{x}}{x-1} \times \frac{1+\sqrt{x}}{1+\sqrt{x}}$$

$$\frac{(1^2 - x)}{(x-1)(1+\sqrt{x})}$$

$$\frac{(1-x)}{(x-1)(1+\sqrt{x})}$$

$$= \frac{-\cancel{(x-1)}}{\cancel{(x-1)}(1+\sqrt{x})}$$

$$= -\frac{1}{(1+\sqrt{1})}$$

$$= -\frac{1}{2}$$

$$c = -\frac{1}{2}$$

Q4) find the value of constant k if possible, that will make the function continous anywhere

$$f(x) = \begin{cases} 7x - 2 & x \le 1 \\ kx^2 & x > 1 \end{cases}$$

$\lim_{x \to 1}$

left

$7(1) - 2$

$= 5$

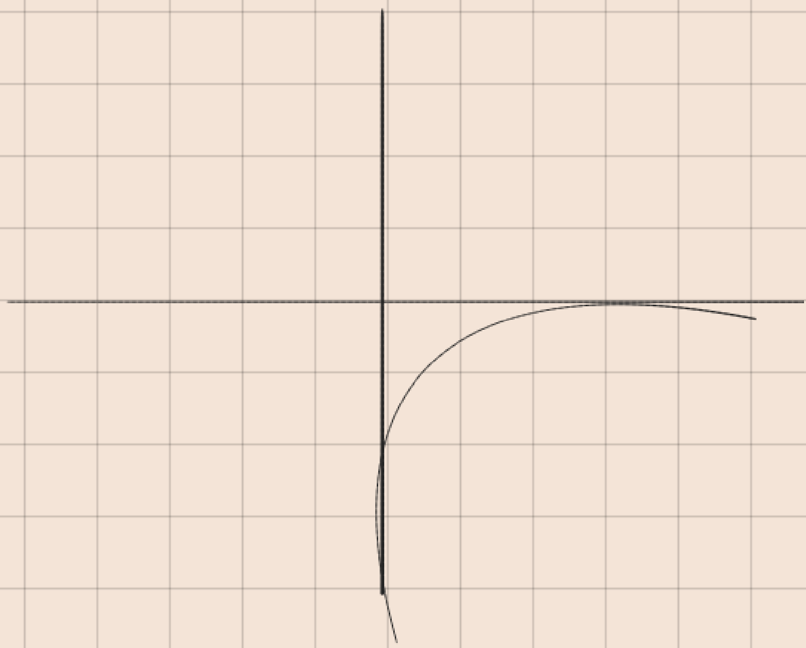
right

$k(1)^2 = 5$

$k = 5$

Functions

Q1) $f(x) = \ln(x-2)$



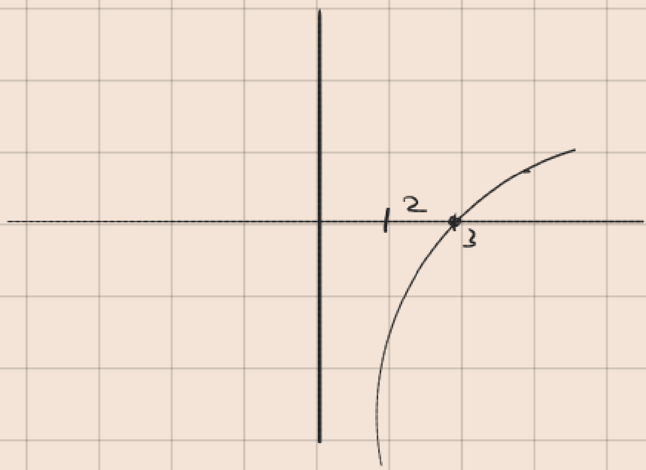
$x - 2 = 0$

$x = 2$

Domain $(2, \infty)$

Range $($

$f(x) = \ln(x-2)$



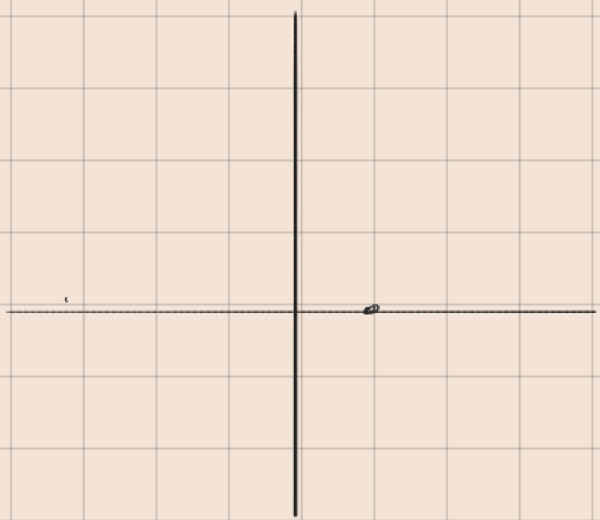
$x-2=0$

$x=2$

$\ln(1)$

$(2, \infty$

$f(x) = e^{x-2}$



$x - 2 = 0$

$x = 2$

$e^{x} - 3$

$x = 0$

$1 - 3$

$= -2$

$f(x) = 2\ln(x+1) - 3$

$=$

$x = 0$

$y = -3$

$x = 1$

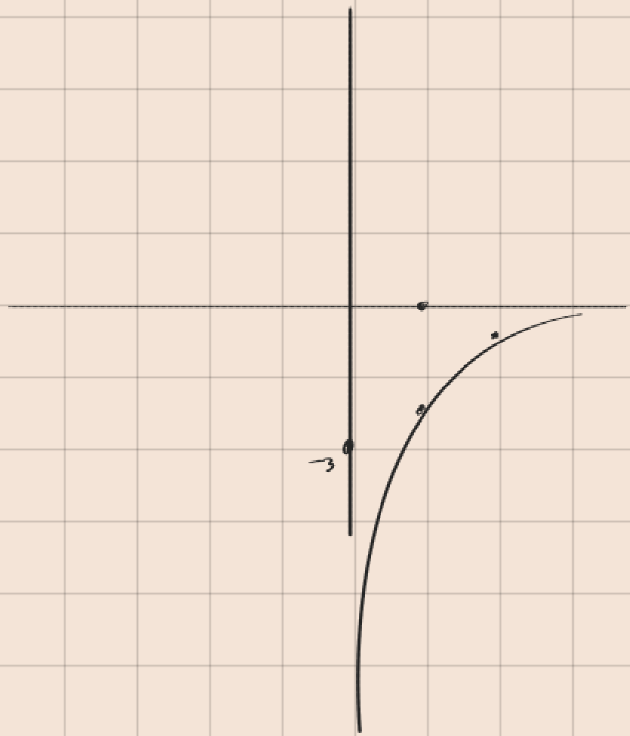
$x = 2$

Domain

$(0, \infty)$

Range

$(-$

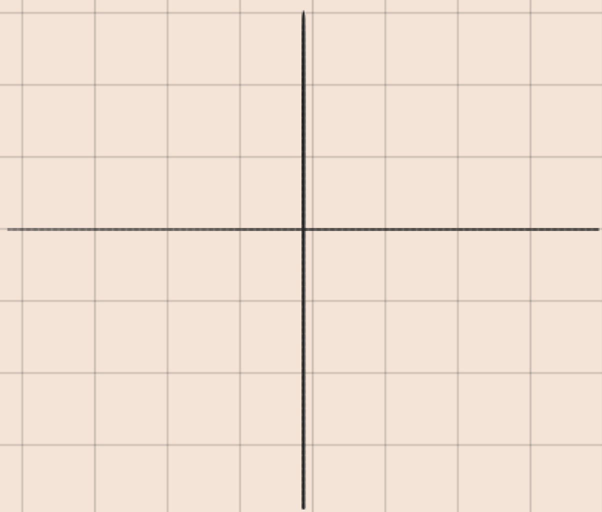


$f(x) = -3x^2 + 2x - 4$

vertex $= -\frac{b}{2a}$

$= -\frac{(2)}{2(-3)}$

$= \frac{1}{3}$



$f(x) = 6(x+1)^2 - 11$

$= 6(x^2 + 2x + 1) - 11$

$= 6x^2 + 12x + 6 - 11$

$= 6x^2 + 12x - 5$

$x = \dfrac{-(12)}{2(6)}$

$x = -1 \qquad y = -11$

vertex $(-1, -11)$

Composite Functions

$f(x) = x^2 - 3$ $\qquad$ $h(x) = \sqrt{x+1}$

1) $(f \circ h)(x)$

$= (\sqrt{x+1})^2 - 3$

$= x + 1 - 3$

$(f \circ h)(x) = x - 2$

2) $(h \circ f)(x)$

$= \sqrt{(x^2 - 3) + 1}$

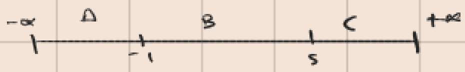
$= \sqrt{x^2 -}$

Rational Inequalities

Q1) $\dfrac{x+1}{x-5} \le 0$

$x+1 = 0 \qquad x-5=0$

$x=-1 \qquad x=5$



$x < -1$	-2	$\frac{1}{7} \le 0$	$-$
$-1 < x < 5$	0	$-\frac{1}{5} \le 0$	$+$
$x > 5$	6	$7 \le 0$	$-$

$x \in [-1, 5)$

Q) $\dfrac{x^2+4x+3}{x-1} > 0$

$\dfrac{x^2+(3+1)x+3}{x-1}$

$\dfrac{x^2+3x+x+3}{x-1}$

$\dfrac{(x+3)(x+1)}{x-1} > 0$

$x=-3 \qquad x=-1 \qquad x=1$

Ans = $x \in (-3,-1)$
$\cup\ (1, \infty)$

$x < -3$	-4	$-\frac{3}{5} > 0$	$-$
$-3 < x < -1$	-2	$\frac{1}{3} > 0$	$+$
$-1 < x < 1$	0	$-3 > 0$	$-$
$x > 1$	2	$15 > 0$	$+$

e) $\dfrac{x^2-16}{(x-1)^2} < 0$

$\dfrac{(x-4)(x+4)}{(x-1)^2} < 0$

$x = 4 \qquad x \; -4 \qquad (x-1)^2 = 0$

$x-1=0$

$x=1$

$x < -4$	-5	$\frac{1}{4} < 0$	$-$
$-4 < x < 1$	0	$-16 < 0$	$+$
$1 < x < 4$	2	$-12 < 0$	$+$
$x > 4$	5	$\frac{9}{16} < 0$	$-$

$x \in (-4, 1) \cup (1, 4)$

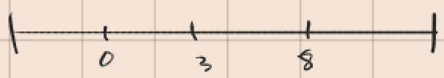
o)

$$\frac{x-8}{x} \leq 3-x$$

$$\frac{x-8}{x(3-x)} \leq 0$$

$$\frac{x-8}{x(3-x)} \leq 0$$

$x=8$ $\quad$ $x=0$ $\quad$ $x=3$



$x<0$	-1	$\frac{9}{4} \leq 0$	$-$
$0<x<3$	2	$-3 \leq 0$	$+$
$3<x<8$	5	$\frac{3}{10}$	$-$
$x>8$	9	$-\frac{1}{54}$	$+$

$$x \in (-\infty, 0) \cup [8, \infty)$$

Q) $\frac{x-8}{x} \le x-3$

$\frac{x-8}{x} - (3-x)$

$\frac{x-8-3x+x^2}{x}$

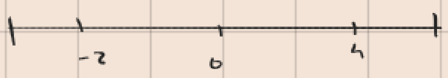
$\frac{x^2-2x-8}{x}$

$\frac{x^2-(4-2)-8)}{x} \le 0$

$\frac{x^2-4x+2x-8}{x} \le 0$

$\frac{x(x-4)+2(x-4)}{x} \le 0$

$x=4 \quad x=-2 \quad x=0$



$x < -2$	-3	$-\frac{7}{3} \le 0$	+
$-2 < x < 0$	-1	5	−
$0 < x < 4$	1	$-9 \le 0$	+
$x > 4$	5		−

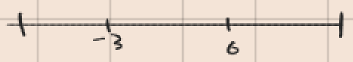
$x \in (-\infty, -2] \cup (0, 4]$

Q) $\dfrac{3x+1}{x+4} \geqslant 1$

$3x+1 = 1$ $\quad$ $x+4 = 1$

$3x = 1-1$ $\quad$ $x = 1-4$

$x = 0$ $\quad$ $x = -3$



$x < -3$	-5	$14 \geqslant 1$	+
$-3 < x < 0$	-1	$-\frac{2}{3} \geqslant 1$	−
$x > 0$	1	$\frac{4}{5} \geqslant 1$	+

$x \in (-\infty, -3) \cup [0, \infty)$

$\theta)$ $|2x-4| < 10$

$2x-4 < \pm 10$

$2x-4 = 10$

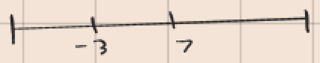
$2x = 14$

$x = 7$

$2x-4 = -10$

$2x = -6$

$x = -3$



$x < -3$	-4	$-$
$-3 < x < 7$	1	$+$
$x > 7$	8	$-$

$\theta)$ $|9m+2| \leq 1$

$9m+2 \leq +1$

$9m = 1-2$

$m = \frac{-1}{9}$

$9m+2 \leq -1$

$m = -1-2$

$m = \frac{-\cancel{81}}{\cancel{9}3}$

$m = \frac{-1}{3}$

$x < -\frac{1}{3}$

$\frac{-1}{3} < x < \frac{-1}{9}$

Q) $|x^2+6x+1| > 8$

$x^2+6x-7>0$

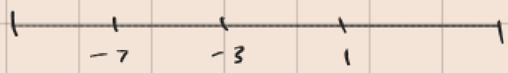
$(x+7)(x-1)>0$

$x=-7$ $\quad$ $x=1$

$x^2+6x+9<0$

$(x+3)(x+3)<0$

$x=-3$, $x=-3$



$2x^2 - 5x - 12 \leq 0$

$2x^2 - (8-3)x - 12 \leq 0$

$2x^2 - 8x + 3x - 12 \leq 0$

$2x(x-4) + 3(x-4) \leq 0$

$(2x+3) \quad (x-4)$

$2x+3=0 \qquad x=4$

$x = -\frac{3}{2} \qquad x = 4$

$-\frac{3}{2} \qquad 4$

$x < -\frac{3}{2}$	-2	$6 \leq 0$	$-$
$-\frac{3}{2} < x < 4$	0	$-12 \leq 0$	$+$
$x > 4$	5	$13 \leq 0$	$-$

$x \in [-\frac{3}{2}, 4]$

$-6 \qquad -3 \qquad 8$

b) $\frac{x+6}{x^2 - 5x - 24} \geq 0$

$\frac{x+6}{x^2 - (8-3)x - 24}$

$\frac{x+6}{x^2 - 8x + 3x - 24}$

$\frac{x+6}{x(x-8) + 3(x-8)}$

$x+6=0 \qquad x-8=0 \qquad x+3=0$

$x=-6 \qquad x=8 \qquad x=-3$

$x < -6$	-7	$-$
$-6 < x < -3$	-4	$+$
$-3 < x < 8$	0	$-$
$x > 8$	9	$+$

$-\frac{1}{60} \geq 0$

$\frac{1}{6} \geq 0$

$-\frac{1}{4} \geq 0$

$\frac{5}{4} \geq 0$

$[-6, -3) \cup (8, +\infty)$

Q $\lim_{x \to 3} \dfrac{x^2 - 3x}{2x^2 - 5x - 3}$

$\dfrac{x^2 - 3x}{2x^2 - (6-1)x - 3}$

$\dfrac{x(x-3)}{2x^2 - 6x + x - 3}$

$\dfrac{x(x-3)}{2x(x-3) + 1(x-3)}$

$\dfrac{x\cancel{(x-3)}}{(2x+1)\cancel{(x-3)}}$

$= \dfrac{x}{2x+1} \qquad = \dfrac{3}{7}$

left

$-x^2 + 4$

$-(3)^2 + 4$

$-9 + 4$

$= -5$

right

$4x + 8$

$4(3) + 8$

$12 + 8$

$\lim_{x \to 3^-} \neq \lim_{x \to 3^+}$

LHL RH

Q $|x-3| - 2 > 6$

$|x-3| > 2$

Q) $-2 \quad |2x+10| - 6 < 0$

$|x-2| - 3 < 0$

$|x-2| < 3$

$x - 2 < +3$

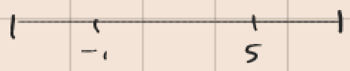
$x < 3 + 2$

$x < 5$

$x - 2 > -3$

$x > -3 + 2$

$x > -1$



$x < -1$	-2	$4 < 3$	$-$
$-1 < x < 5$	0	$2 < 3$	$+1$
$x > 5$	6	$4 < 3$	$-$

$(-1, 5)$

$f(x) = \frac{|x|}{x} \quad \lim_{x \to 0}$

1) $p \wedge q$

2) $q \to r$

3) $t \to s$

4) $\neg t \to u$

5) $s \vee u \to w$

from 1 from 3

6) q

from 6 and 2

7) r

Question 2 [CLO-2] **Marks: 05**

Apply the inference rules and construct a valid argument for the conclusion, and explain why these rules will satisfy premises.

Premises:

1. "The sky is covered with dark clouds and it looks like it might rain heavily so then the outdoor concert will be postponed to another day."
2. "If the outdoor concert is postponed, then the band will instead perform an indoor show at the community hall."
3. "Attending the indoor show is only possible if tickets are available at the box office on the day of the performance."
4. "If tickets are not available at the box office, then we will stay home and watch a live stream of the performance online."
5. "Whenever we either attend the indoor show or watch the live stream at home, we will finish the evening feeling entertained and satisfied."

Conclusion: "We will finish the evening feeling entertained and satisfied."

1) $p \wedge q$

2) $p \rightarrow r$

3) $s \rightarrow r$

4) $\neg s \rightarrow t$

5) $r \vee p \rightarrow u$